

Name .....

Roll No. ....



# ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY, PALAI.

(An ISO 9001:2015 Certified College)

B. TECH DEGREE FIRST INTERNAL TEST- NOVEMBER 2022

FIFTH SEMESTER – COMMON TO ALL BRANCHES

MCN 301 DISASTER MANAGEMENT

Time: 2 Hours

Max.Marks: 50

## PART A

Answer all Questions (3x5=15 marks)

| Qn. No. | Questions                                                                                           | Marks | K level | CO & KL |
|---------|-----------------------------------------------------------------------------------------------------|-------|---------|---------|
| 1       | Explain the significance of the Ozone layer in protecting harmful radiations                        | 3     | K1      | CO1&K2  |
| 2       | What is hazard ? List different types of hazards?                                                   | 3     | K1      | CO1&K2  |
| 3       | Write a brief note on global warming and its impacts.                                               | 3     | K1      | CO2&K2  |
| 4       | State the principle of qualitative risk assessment and the method of expressing risk qualitatively. | 3     | K2      | CO2&K2  |
| 5       | Explain briefly the concepts of 'disaster risk'                                                     | 3     | K1      | CO3&K2  |

## PART B

Answer five questions (7 x 5 = 35 Marks)

Module 1: Qn. No 6,7 and 8 – Answer any 2

Module 2: Qn. No 9,10 and 11 – Answer any 2

Module 3: Qn. No 12 is mandatory

| Qn. No. | Questions                                                                                                                        | Marks | K Level | CO & KL  |
|---------|----------------------------------------------------------------------------------------------------------------------------------|-------|---------|----------|
| 6       | Explain the different layers of atmosphere based on temperature variation in detail.                                             | 7     | K1      | CO1 & K2 |
| 7       | What are the different types of soil and also explain the functions of soil ?                                                    | 7     | K1      | CO1 & K2 |
| 8       | Compare Hazard and Risk in the context of Disaster Management with examples                                                      | 7     | K2      | CO1 & K2 |
| 9       | Explain Disaster Risk assessment and the different ways to express Disaster Risk.                                                | 7     | K1      | CO2 & K2 |
| 10      | Explain the four different types of vulnerability. List any four socio-economic indicators of human capital as livelihood asset. | 7     | K1      | CO2 & K2 |
| 11      | Outline the two major physical vulnerability assessment approaches.                                                              | 7     | K2      | CO2&K2   |
| 12      | Explain the core elements of disaster risk management                                                                            | 7     | K1      | CO3&K2   |

Name .....

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# ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY, PALAI.

(An ISO 9001:2015 Certified College)

## B. TECH DEGREE SECOND INTERNAL TEST- DECEMBER 2022

### FIFTH SEMESTER – COMMON TO ALL BRANCHES

### MCN 301 DISASTER MANAGEMENT

Time: 2 Hours

Max Marks: 50

#### PART A

Answer all Questions (3x5=15 marks)

| Qn. No. | Questions                                                                                                   | Marks | K level | CO & KL |
|---------|-------------------------------------------------------------------------------------------------------------|-------|---------|---------|
| 1       | What is meant by Disaster Relief? Mention the names of any two International Disaster Relief Organizations. | 3     | K1      | CO3&K2  |
| 2       | Explain briefly the three basic forms of stakeholders                                                       | 3     | K1      | CO5&K2  |
| 3       | Define Disaster Mitigation? Briefly explain the types of Disaster Mitigation measures                       | 3     | K1      | CO4&K2  |
| 4       | What is NDMA? Name any four functions of it.                                                                | 3     | K2      | CO6&K2  |
| 5       | Explain the relevance of Disaster Management Act 2005.                                                      | 3     | K1      | CO6&K2  |

#### PART B

Answer five questions (7 x 5 = 35 Marks)

Module 4: Qn. No 6,7 and 8 – Answer any 2

Module 5: Qn. No 9,10 and 11 – Answer any 2

Module 3: Qn. No 12 is mandatory

| Qn. No. | Questions                                                                                               | Marks | K Level | CO & KL  |
|---------|---------------------------------------------------------------------------------------------------------|-------|---------|----------|
| 6       | (a) How Capacity assessment is done and strengthened?                                                   | 4     | K1      | CO4 & K2 |
|         | (b) Explain any 3 effective ways of promoting Stakeholder participation in disaster risk reduction      | 3     |         | CO5 & K2 |
| 7       | (a) Define Disaster Communication. What are the barriers to Effective Communication                     | 4     | K2      | CO4 & K2 |
|         | (b) Explain the importance of communication in disaster risk reduction.                                 | 3     |         | CO5 & K2 |
| 8       | (a) Explain Capacity building in the context of Disaster management                                     | 4     | K2      | CO4 & K2 |
|         | (b) Explain the different levels of Stakeholder participation in the context of Disaster Risk Reduction | 3     |         | CO5 & K2 |

|    |                                                                                                                                         |        |    |          |
|----|-----------------------------------------------------------------------------------------------------------------------------------------|--------|----|----------|
| 9  | (a) Explain the common natural disaster types in India.<br>(b) Elaborate on man made disasters in India with case studies.              | 3<br>4 | K1 | CO6 & K2 |
| 10 | (a) List any three objectives of National Policy on Disaster Management<br>(b) How Tsunami are created                                  | 3<br>4 | K1 | CO6 & K2 |
| 11 | Explain guiding principles of Sendai Framework for disaster risk reduction                                                              | 7      | K2 | CO6 & K2 |
| 12 | (a) Explain the primary objectives of Disaster Mitigation<br>(b) Explain the components of comprehensive disaster preparedness strategy | 3<br>4 | K2 | CO3&K2   |

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Reg No.: \_\_\_\_\_

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**APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY**

Fifth Semester B.Tech Degree Regular and Supplementary Examination December 2022 (2019 Scheme)

**Course Code: MCN 301****Course Name: DISASTER MANAGEMENT**

Max. Marks: 100

Duration: 3 Hours

**PART A***(Answer all questions; each question carries 3 marks)*

Marks

- |      |                                                                               |   |
|------|-------------------------------------------------------------------------------|---|
| ✓ 1  | Differentiate between acceptable risk and residual risk.                      | 3 |
| ✓ 2  | State the composition of lithosphere?                                         | 3 |
| ✓ 3  | Explain physical vulnerability and ecological vulnerability.                  | 3 |
| ✓ 4  | Define disaster deficit index.                                                | 3 |
| ✓ 5  | Explain psychosocial support related with disaster response.                  | 3 |
| ✓ 6  | List any six public health services required in responding to disasters.      | 3 |
| ✓ 7  | Explain pyramid scheme related with participatory stakeholder engagement.     | 3 |
| ✓ 8  | Define the terms counselling and crisis counselling.                          | 3 |
| ✓ 9  | Define man-made disasters and list examples from our country.                 | 3 |
| ✓ 10 | Explain the role of National Institute of Disaster Management in our country. | 3 |

**PART B***(Answer one full question from each module, each question carries 14 marks)***Module -1**

- |    |                                                                                                               |   |
|----|---------------------------------------------------------------------------------------------------------------|---|
| 11 | a) Illustrate with diagram the layers of earth's atmosphere.                                                  | 8 |
|    | b) State and explain the classification of rocks.                                                             | 6 |
| 12 | a) State and explain the features of hydrosphere and biosphere in detail.                                     | 8 |
|    | b) Explain the terms resilience, recovery and early warning systems with respect to disaster risk management. | 6 |

**Module -2**

- |    |                                                                                                    |   |
|----|----------------------------------------------------------------------------------------------------|---|
| 13 | a) Explain the applications of hazard maps.                                                        | 8 |
|    | b) Explain the methods of representing vulnerability.                                              | 6 |
| 14 | a) Explain the method of expressing population risk.                                               | 5 |
|    | b) List the components of risk assessment. Explain the contemporary approaches to risk assessment. | 9 |

**Module -3**

- 15 a) State and explain the types of disaster mitigation measures. 4  
 b) Identify the factors that determine the nature of disaster response and explain. 10
- 16 a) State and explain the types of disaster preparedness. 6  
 b) Identify the standard operating procedures to be followed during a disaster stage and explain. 8

**Module -4**

- 17 a) State and explain the basic principles of participatory rural appraisal tools. 5  
 b) Explain the characteristics of effective crisis counsellors. State the advantages of crisis counselling. 9
- 18 a) State and explain the steps for effective communication. 8  
 b) Identify the barriers to effective communication. 6

**Module -5**

- 19 a) List the global targets of Sendai framework and explain. 7  
 b) Explain the role, composition and responsibilities of National Disaster Management Authority. 7
- 20 a) List the guiding principles of Sendai framework and explain. 10  
 b) State the composition and role of National Executive Committee related with the National Policy on Disaster Management. 4

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